

Outcomes After Proton Beam Therapy for Large Choroidal Melanomas in 492 Patients



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- **PURPOSE:** To evaluate proton beam therapy (PBT) as a means to preserve the eye and spare some vision while not deteriorating survival in patients with large choroidal melanomas.

- **DESIGN:** This is a retrospective, consecutive cohort study of patients with T3-4 choroidal melanomas according to the 7th edition of the American Joint Cancer Classification treated with PBT over a 24-year period.

- **RESULTS:** A total of 492 patients were included. Mean (range) tumor thickness and diameter were 8.77 (2–15) mm and 14.91 (7–24.1) mm, respectively. Mean macular and optic disc distance were 4.56 (0–19.9) mm and 4.59 (0–22.1) mm, respectively. Mean follow-up was 61.9 months. Rates of neovascular glaucoma (NVG) and enucleation (mainly for local recurrence or NVG) were 27.0% and 19.5%, respectively. Enucleation rates decreased over time. The 5-year local control was 94%. Mean baseline visual acuity was 20/63, and visual acuity $\geq 20/200$ was preserved in 20% of patients. At 5 years, 25% of T3 patients presented with metastasis; overall and specific survival rates were 65% and 75%, respectively.

- **CONCLUSION:** Local control after PBT remained good with increasingly manageable complications and fewer secondary enucleations over time for these large melanomas. As PBT does not seem to deteriorate survival in these patients having a high risk of metastasis, PBT may be considered as a safe and efficient alternative to enucleation in patients with large choroidal melanoma, and may help to spare some vision. (*Am J Ophthalmol* 2016;165:78–87. © 2016 Elsevier Inc. All rights reserved.)

UVEAL MELANOMA HAS AN INCIDENCE OF 5–6 cases per million per year.^{1,2} It is the most common primary intraocular malignancy and it is also the most common noncutaneous melanoma.³

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Historically, enucleation has been the gold standard in the treatment of malignant intraocular tumors. Starting in the seventies, conservative approaches, such as ophthalmic plaque brachytherapy,⁴ have been developed for small to medium-sized melanomas. Further, a prospective and Collaborative Ocular Melanoma Study (COMS) suggested, in 2006, that small to medium-sized choroidal melanomas could be treated by enucleation or conservative treatment using brachytherapy, with similar survival and metastasis-free survival rates with the 2 treatment modalities.⁵ At that time, the American Brachytherapy Society declared that patients with a clinical diagnosis of medium-sized choroidal melanoma (between 2.5 and 10 mm in height and < 16 mm basal diameter) were candidates for episcleral plaques.⁶ However, with plaque radiotherapy, the higher the tumor, the higher the scleral dose, with difficulties to deliver a sufficient dose to the apex of the tumor for high tumors, depending on the radionuclide used. Moreover, treating tumors close to the optic nerve can also be challenging with plaque brachytherapy.

Another conservative approach uses proton beam therapy (PBT).⁷ A COMS trial compared external beam radiotherapy followed by enucleation with enucleation alone for patients with large choroidal melanomas and found similar 5- and 10-year survival rates.^{8,9} Additionally, the use of PBT has some advantages compared to brachytherapy. PBT allows for the treatment of even higher tumors and tumors closer from the optic nerve than brachytherapy, using a spread-out Bragg peak, which results in a steep distal dose fall-off beyond the tumor. While PBT may be associated with more anterior segment complications, preliminary experiences have shown interesting eye retention rates following PBT.¹⁰ Moreover, despite less favorable selection biases in irradiated vs enucleated patients, a retrospective and comparative study corroborated these findings in showing that cumulative all-cause mortality, melanoma-related mortality, and metastasis-free survival were not statistically different between the two.¹¹ Also, eye retention and best-corrected visual acuity (BCVA) over 20/200 were obtained in about 75% of patients.

One particular challenge relates to large uveal tumors, as they may be associated with an increased risk of local failure with conservative therapies.^{12,13} Additionally, large irradiated volumes at very high doses are at higher risk for complications that may ultimately result in secondary enucleation.¹⁴ Another area of concern is the intrinsically

high risk of liver metastasis in this population.¹⁵ Altogether, the management of large uveal melanomas is challenging.

In consideration of the trend toward using PBT for ever larger uveal melanomas, we evaluated its validity in treating T3-T4 uveal melanomas with respect to survival, metastasis-free survival, local complications, and enucleation.

METHODS

THIS IS AN OBSERVATIONAL MONOCENTRIC CONSECUTIVE cohort study with national review board–approved prospective data collection for patients with a large choroidal melanoma treated by proton beam therapy at the cyclotron of the Centre Antoine Lacassagne cancer center (Nice, France) from June 1991 to June 2015.

The 2009 edition of the American Joint Committee on Cancer (AJCC) defines ciliochoroidal melanoma T stage by tumor height, basal diameter, extension to the ciliary body, and/or presence of extrascleral extent.¹⁶ Patients with AJCC tumor (T) stage 3–4 were included according to national guidelines if they had tumors smaller than 18 mm in maximum diameter and thickness inferior to 12 mm, extrascleral extension inferior to 2 mm, tumor/eyeball volume below 40%, and absence of unmanageable glaucoma.

The patient database included data on demographics, laterality (right or left), location of the tumor (pre-equatorial, equatorial, or retroequatorial), tumor measurements, distance between posterior tumor margins to the optic disc and macula (mm), presence of inflammation, increased intraocular pressure, intravitreal hemorrhage, retinal detachment, iris root involvement, extrascleral extension, vorticos vein involvement, and BCVA at baseline and on follow-up. BCVA was assessed in the Snellen system and converted into a logMAR scale using the formula $-\log(\text{visual acuity})$ for statistical analysis. No light perception, light perception only, hand motion, and count fingers were considered as 3, 2.5, 2.3, and 1.9 logMAR, respectively. Tumor basal diameter and thickness were measured by ultrasonography and indirect ophthalmoscopy with fundus photographs, peroperative eye transillumination, and fluorescein angiography were also performed for accurate tumor characterization. Optical coherence tomography and ultrabiomicroscopy were performed based on tumor location and the ophthalmologist's appreciation. Metastatic diagnostic evaluation consisted of liver ultrasonography or computed tomography (CT) scan and liver function. All patients were informed of the risks and potential benefits related to proton therapy and enucleation. We evaluated the complications of PBT such as cataract, neovascular glaucoma

(NVG), retinopathy, neuropathy, intravitreal hemorrhage, retinal detachment, and enucleation.

- **PROTON BEAM THERAPY PLANNING:** The method has been described elsewhere.⁷ In brief, the method involved surgical insertion of 4 tantalum fiducials, sewed onto the sclera around the transilluminated tumor by the onco-ophthalmologist under a short general anesthesia. The dimensions of the tumor were noted, along with the tantalum fiducial–limbus and interfiducial distances. CT-based PBT planning was performed 2 weeks later. Tridimensional eye modeling and tumor/organs at risk delineation was generated using the eye-dedicated treatment planning system, according to the information provided by fiducial placement, retinography, and angiographies. The total dose delivered was 60 cobalt gray equivalent in 4 fractions, which is the usual protocol used in Europe.

- **PATIENT FOLLOW-UP:** Patients were regularly followed by their onco-ophthalmologist every 6 months for 2 years, and yearly thereafter. A retinal photography or angiography, measurement of visual acuity, and ultrasound B for the evaluation of tumor regression¹⁷ were performed. Patients had an initial monthly follow-up by their local ophthalmologist to assess complications. Patients were addressed to the onco-ophthalmologist in case of occurrence of any radiation-related ocular events. After the first year, patients underwent interval biannual visits with their local ophthalmologist. Metastatic screening was performed every 6 months for 5 years with liver ultrasound and liver function tests.

- **STATISTICS:** Statistical analyses were performed using SPSS 22.0 (SPSS Inc, Chicago, Illinois, USA) statistical package for Macintosh. Continuous data are presented as the mean $\pm$ standard deviation (SD) while categorical variables were presented as frequencies and percentages. Continuous variables were compared using Student *t* test while categorical variables were compared using the χ^2 test. Univariate and multivariate Cox proportional hazards regression analyses were performed to relate clinical and ultrasound parameters to the occurrence of NVG, enucleation, overall and specific survival, and occurrence of metastasis. Event rates for the overall, specific, and free-of-NVG survival were plotted in Kaplan-Meier curves, and the log-rank test was used to compare overall and specific survival across T3 and T4 stages. All statistical tests were 2-sided, and a *P* value of $<.05$ was considered statistically significant.

RESULTS

- **DESCRIPTION OF PATIENT POPULATION:** During a 24-year accrual period, 492 out of 1460 patients (33.7%)

TABLE 1. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Patient Demographics and Tumor Characteristics

	Number	(%)
Sex		
Male	258	(52.8)
Female	234	(47.6)
Eye		
Right	238	(48.4)
Left	253	(51.4)
Pre-existing nevus	2	(0.4)
Tumor location		
Pre-equatorial	107	(21.7)
Equatorial	274	(55.7)
Retroequatorial	108	(22.0)
Initial inflammation	59	(12.0)
Initial intravitreal hemorrhage	61	(12.4)
Initial presence of metastasis	8	(1.6)
Extrascleral extension	22	(4.5)
Vorticosse vein involvement	30	(6.1)
Ciliary body involvement	205	(41.7)
Iris root involvement	38	(7.7)
T stage		
T3	416	(84.6)
T4	75	(15.2)
	Mean	Range
Age (y)	62.4	19–89
BCVA before treatment (Snellen scale)	20/63	20/1000–20/20
Tumor thickness (mm)	8.77	2–15
Tumor diameter (mm)	14.91	7–24.1
Tumor volume (cc)	0.92	0.01–19

BCVA = best-corrected visual acuity.

have been treated by proton beam therapy for a large choroidal melanoma. Of these, 416 patients were classified as T3 (84.6%) and 75 as T4 (15.2%). Mean follow-up was 61.9 months (SD 47.8 months). Qualitative and quantitative tumor characteristics are shown in [Tables 1 and 2](#).

Two hundred and fifty-eight patients were male (52.4%) and 234 were female (47.6%). Mean age at diagnosis was 62.4 years (19–89). Both eyes were equally involved (48.5% in the right eye and 51.5% in the left eye). Mean tumor thickness and diameter were 8.77 (2–15) mm and 14.91 (7–24.1) mm, respectively. Mean tumor volume was 0.92 cc (0.01–19). Mean macular and optic disc distance were 4.56 (0–19.9) mm and 4.59 (0–22.1) mm, respectively. Tumor location was pre-equatorial in 21.7%, equatorial in 55.7%, and retro-equatorial in 22.0%. Mean BCVA on the Snellen scale was 20/63 before treatment and 20/160 at last follow-up visit. At last visit, BCVA was inferior to 20/200 for 80.2% of the patients and superior to 20/40 in 8.5%. Among the nonenucleated patients, 76.3% of patients could at least experience light

TABLE 2. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Tumor Location

Tumor Location	T3 (N = 416) Number (%)	T4 (N = 75) Number (%)	P (χ^2)
Juxtapapillary	77 (18.5%)	15 (20%)	.35
Parapapillary	174 (41.8%)	29 (38.7%)	.25
Ciliary body involvement	153 (36.8%)	52 (69.3%)	<.001
Macular involvement	96 (23.2%)	19 (25.3%)	.08

Juxtapapillary: ≤ 0 mm from the optic disc.

Parapapillary: ≤ 3 mm from the optic disc.

perception ([Figure 1](#)). Comparing final to baseline BCVA, 91.4% of patients had lost ≥ 0.3 logMAR of visual acuity and 8.6% had steady or better visual acuity. Of those patients losing VA, 30.4%, 56%, 72.2%, and 91.4% had lost more than 0.3 logMAR of VA within the first 2 years, between 2 and 5 years, between 5 and 10 years, and between 10 and 20 years, respectively ([Figure 2](#)). Of the 492 large choroidal melanoma patients, 12.4% had an intravitreal hemorrhage, 62.8% had a retinal detachment, 4.5% had an extrascleral extension, 41.7% had a ciliary body extension, and 7.7 % had iris root involvement at baseline. Of these, 1.6% had liver metastasis at diagnosis of ocular disease.

• **EVENTS DURING FOLLOW-UP:** Anterior and posterior segment complications after proton beam therapy are displayed in [Table 3](#). In this T3-T4 population, 27.0% (119 T3 and 14 T4) of the patients had NVG. Of these, 40.6% patients were enucleated. The mean delay between proton beam therapy and NVG was 21.65 months (SD 26.13 months). Risk factors for NVG by univariate and multivariate Cox proportional hazard regression analyses were ([Table 4](#)) distance to the optic disc ($P < .01$) and macula ($P < .01$), presence of a retinal detachment at last visit ($P < .01$), initial tumor thickness ($P < .01$), initial tumor diameter ($P = .01$), initial presence of an intravitreal hemorrhage ($P < .01$), and local recurrence ($P < .01$). Overall, 19.5% of the patients were enucleated. Reasons for enucleation were NVG (30.2%), phthisis (12.5%), relapse (32.3%), and unreliable surveillance (24.8%). The mean delay between proton beam therapy and enucleation was 35.9 months (SD 43.9 months). Risk factors for enucleation on univariate and multivariate analyses ([Table 5](#)) were local recurrence ($P < .01$), NVG ($P < .01$), radiation-induced retinopathy ($P = .02$), presence of a retinal detachment after treatment ($P = .03$), TNM stage ($P < .01$), and initial tumor thickness ($P < .01$). Owing to changes in the management of complications with the use of intravitreal VEGF inhibitors in 2008, we evaluated NGV and enucleation rates before and after 2008. The NVG rate increased from 21.6% to 36.7% while

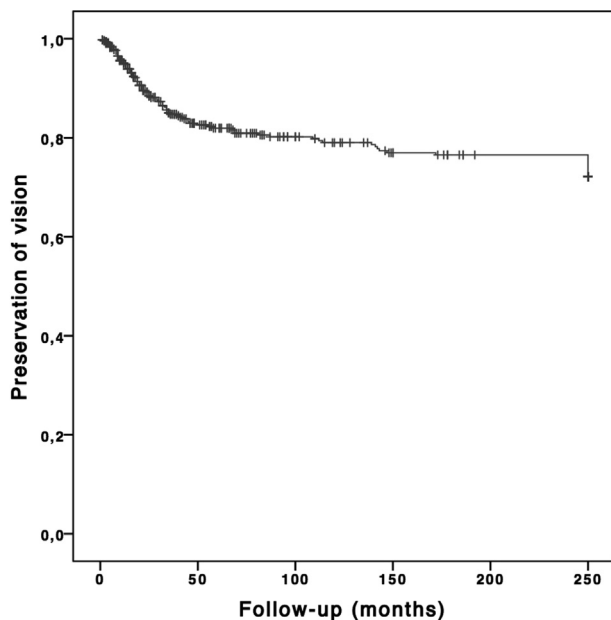


FIGURE 1. Proton beam therapy for large choroidal melanomas in 492 consecutive cases: survival with light perception or better in large choroidal melanoma patients by Kaplan-Meier analysis.

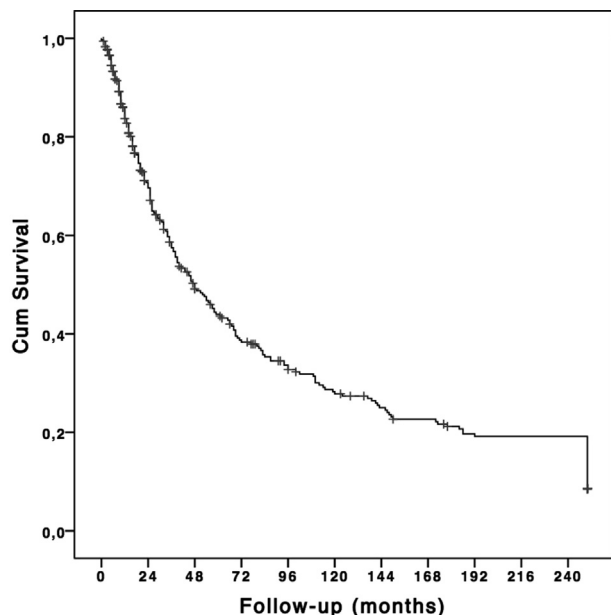


FIGURE 2. Proton beam therapy for large choroidal melanomas in 492 consecutive cases: Kaplan-Meier estimate of patients free of visual deterioration (patients experiencing ≥ 0.3 logMAR difference in visual acuity).

enucleation rates decreased from 38.2% to 20.0% during the same periods of time ($P < .01$), regardless of tumor size.

In our population, 39.8% developed cataract, 23.4% presented with a radiation retinopathy, 8.3% had a

TABLE 3. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Anterior and Posterior Segment Complications

Complications	Total N (%)	T3 (N = 416) N (%)	T4 (N = 75) N (%)
Cataract	196 (39.8)	164 (39.4)	32 (42.7)
Neovascular glaucoma	133 (27.0)	119 (28.6)	14 (18.7)
Retinopathy	115 (23.4)	102 (24.5)	13 (17.3)
Neuropathy	41 (8.3)	38 (9.1)	3 (4.0)
Intravitreal hemorrhage	84 (17.1)	78 (18.8)	6 (8.0)
Lipidid exudation	57 (11.6)	52 (12.5)	5 (6.7)
Dry eye syndrome	73 (14.8)	65 (15.6)	8 (10.7)
Lacrimal stenosis	16 (3.3)	16 (3.8)	0 (0.0)
Alopecia	53 (10.8)	48 (11.5)	5 (6.7)
Dermatitis	29 (5.9)	24 (5.8)	5 (6.7)
Enucleation	96 (19.5)	74 (17.8)	22 (29.3)
Retinal detachment		Before Treatment N (%)	After Treatment N (%)
0 = none	0	183 (37.2)	276 (56.1)
1 = peritumoral	1	87 (17.7)	26 (5.3)
2 = 1 or 2 quadrants	2	204 (41.5)	56 (11.4)
3 = 3 or 4 quadrants	3	13 (2.6)	23 (4.7)

radiation neuropathy, and 17.1% had an intravitreal hemorrhage following proton beam therapy. Other complications included dry eye syndrome (14.8%), lacrimal stenosis (3.3%), dermatitis (5.9%), and alopecia (10.8%).

• **LOCAL TUMOR CONTROL:** Local control was achieved in 94% of the patients at year 5. Mean time to local relapse was 29.0 months (SD 27.40). Local relapses were treated with proton reirradiation (24.1%), enucleation (65.5%), or laser (10.3%). Initial tumor thickness ($P < .01$) was a risk factor for local relapse on univariate and multivariate analyses.

• **SURVIVAL AND METASTASIS:** Median overall survival in the combined T3 and T4 population was 126.6 months (range 116.7–136.4 months for the patients who experienced the event). The Kaplan-Meier estimate for overall survival was 68% and 52% at 5 years and 54% and 40% at 10 years in the T3 and T4 patients, respectively (Figure 3 and $P = .01$). The Kaplan-Meier estimate for specific survival was 77% and 62% at 5 years and 71% and 59% at 10 years in the T3 and T4 patients, respectively (Figure 4 and $P = .02$). In multivariate analysis, risk factors for poorer overall survival (Table 6) were an older age ($P < .01$), juxtapapillary location (tumor abutting the optic disc) ($P = .02$), and occurrence of metastasis during follow-up ($P < .01$). Using Kaplan-Meier estimates, metastases were detected in 25.0% of the patients at 5 years, 25.0%,

TABLE 4. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Factors Predictive for Neovascular Glaucoma

	Univariate Analysis			Multivariate Analysis		
	HR	95% CI	P Value	HR	95% CI	P Value
Tumor thickness	1.15	1.07–1.24	<.01	1.16	1.06–1.26	<.01
Local recurrence	1.31	1.08–1.58	<.01	1.33	1.06–1.66	.02
Retinal detachment after treatment	1.57	1.42–1.73	<.01	1.46	1.31–1.63	<.01
Initial intravitreal hemorrhage	2.99	2.10–4.29	<.01	2.31	1.57–3.39	<.01

CI = confidence interval; HR = hazard ratio.

TABLE 5. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Factors Predictive for Enucleation

	Univariate Analysis			Multivariate Analysis		
	HR	95% CI	P Value	HR	95% CI	P Value
Stage T of 7th TNM	1.88	1.17–3.02	.01	2.27	1.29–3.99	<.01
Tumor thickness	1.35	1.23–1.50	<.01	1.28	1.15–1.53	<.01
Local recurrence	1.48	1.25–1.77	<.01	1.74	1.45–2.08	<.01
Neovascular glaucoma	14.84	4.24–51.92	<.01	28.54	3.30–246.72	<.01
Radiation-induced retinopathy	3.37	1.75–6.50	<.01	2.52	1.13–5.63	.02
Retinal detachment after treatment	11.35	3.29–39.14	<.01	10.17	1.19–87.33	.03

10 years, and 30.0% at 15 years in T3 patients, and 35.0% at 5 or 10 years and 43.0% at 15 years in T4 patients. The mean time to metastasis was 33.3 months (SD 36.5). Risk factors for the development of metastases were juxtapapillary location ($P < .01$), extrascleral extension ($P = .046$), and ciliary body extension ($P < .01$) (Table 7). Genomics are not available, as they have only been performed since 2013.

DISCUSSION

LARGE CHOROIDAL MELANOMAS, AS DEFINED BY THEIR thickness, diameter, and extrascleral extension in the AJCC or COMS classifications, have long been treated with enucleation. In consideration of the worldwide trend toward more conservative treatments in recent years, we evaluated our use of proton beam therapy as a primary eyeball-sparing treatment in large choroidal melanoma patients. We show that this strategy not only allows for eye preservation without deteriorating oncologic outcomes in a substantial proportion of patients but also can be used as a means of sparing some vision with increasingly manageable complications.

Enucleation used to be the gold standard in the treatment of large choroidal melanoma. However, in 1978,

Zimmerman and associates cast doubt on the benefits of enucleation and suggested that enucleation may actually promote tumor metastasis.¹⁸ Recent preclinical data in murine models suggested that proton therapy may indeed reduce the risk.¹⁹ The COMS clinical study did not show any increased risk of metastases with preoperative irradiation for large melanomas compared with enucleation.^{8,9} Knowing that median survival after the diagnosis of liver metastases is about 20 months, the relevance of eye removal and the detrimental effects of mutilation are questionable. Thus, large melanoma patients may benefit from a conservative approach provided that complications are manageable and do not significantly deteriorate the quality of life. With improved management of complications using various medical and surgical intraocular treatments in recent years,^{20–22} it is likely that local complications following proton beam therapy will be better managed.

The definition of large choroidal melanoma used in the studies available is very different. Mosci and associates¹¹ performed a comparative study between 132 patients undergoing enucleation and proton therapy for large choroidal melanomas. With the limitation of biases inherent to retrospective studies, there was no difference in regard to survival after proton beam therapy and enucleation. While Mosci, Semenova and Finger,¹⁵ and we used the latest AJCC classification, Shields and

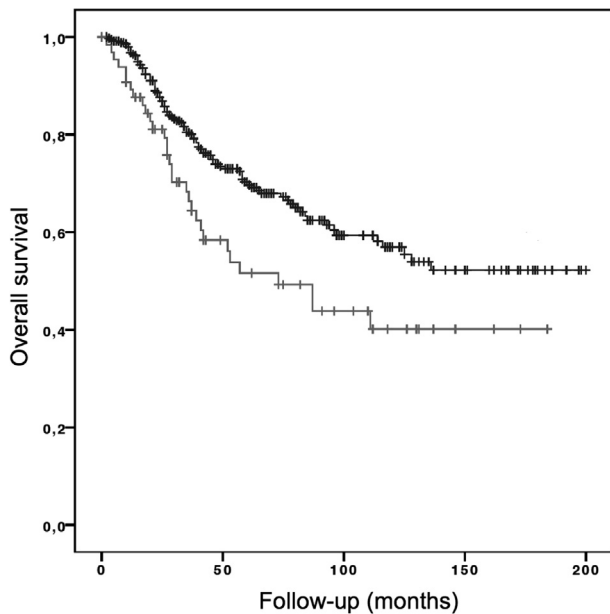


FIGURE 3. Kaplan-Meier curves for overall survival in patients with large choroidal melanoma after proton beam therapy. Separate results for patients with a T3 choroidal melanoma (black) and a T4 choroidal melanoma (gray).

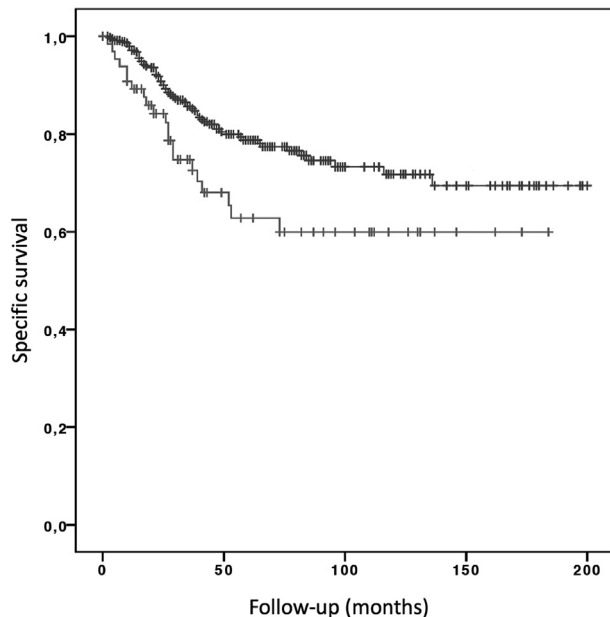


FIGURE 4. Kaplan-Meier curves for specific survival in patients with large choroidal melanoma after proton beam therapy. Separate results for patients with a T3 choroidal melanoma (black) and a T4 choroidal melanoma (gray).

associates²³ and Bechrakis and associates²⁴ used tumor thickness only (with an 8 mm and 7.5 mm threshold, respectively) as a definition of large melanomas and Conway and associates²⁵ and Fuss and associates²⁶ used

the COMS definition (thickness ≥ 10 mm or diameter ≥ 20 mm). Fuss and associates²⁶ evaluated medium and large choroidal melanomas. Direct comparisons can thus be difficult. Nevertheless, the AJCC 2009 classification accurately reports tumors of any size and thickness and is now quite consensual.^{16,27,28}

With 492 patients treated over 24 years, our study is the largest study published to date treating large choroidal melanomas with any modality (brachytherapy or proton beam therapy) of irradiation (Table 8). The COMS study included 1003 patients, but all underwent systematic enucleation either up front or following brachytherapy.⁵ The second-largest is Shields's study²³ with 354 large choroidal melanoma patients treated with brachytherapy over 21 years. Owing to the capacity of proton beam therapy to treat large volumes homogeneously within the whole tumor volume, to treat melanomas near the optic nerve and strengthened by our 25-year experience in such treatments, we could treat all consecutive patients with proton beam therapy. In contrast, brachytherapy can be particularly challenging with respect to plaque placement and tumor coverage in large and/or parapapillary melanomas.^{15,23,24,29,30} The insertion of large plaques includes the necessity to perform extensive ocular muscle sections. It also requires several days of hospitalization. Plaques are removed while tantalum fiducials can be left in place. Tumor coverage is also compromised with plaques in thick and parapapillary melanomas, sometimes requiring additional treatments, which, however, do not appear to fully compensate for the heterogeneous and insufficient dose at the tumor apex. The smaller numbers of patients in the other large melanoma studies might thus reflect more stringent patient selection criteria. In addition, some centers advocate enucleation in patients with large choroidal melanomas with no light perception.¹⁵ However, we believe that eyeball preservation may still be important for cosmesis and social life in these patients, regardless of their vision.

It is common belief that brachytherapy yields fewer severe complications than proton therapy in the ophthalmology community. Brachytherapy has been more ubiquitous (requires cheaper equipments compared to cyclotrons necessary for proton beam therapy) worldwide and has generated about twice the number of publications done with eye proton therapy. However, brachytherapy delivers high scleral doses, which can lead to scleral melting, a complication not present after proton beam therapy. Shields²³ and Semenova's¹⁵ studies using plaque radiotherapy for large choroidal melanomas and ours using proton therapy yielded rates of radiation retinopathy, radiation neuropathy, cataract, and intravitreal hemorrhage of 43.0% and 66.0% vs 23.4%, 22.0% and 51.0% vs 8.3%, 66.0% and 36.0% vs 39.8%, and 23.0% and 13.0% vs 17.1%, respectively. Conway and associates' study²⁵ of extra-large melanomas (thickness ≥ 10 mm or diameter ≥ 20 mm) using PBT found rates of radiation-induced

TABLE 6. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Factors Predictive for Overall Survival

	Univariate Analysis			Multivariate Analysis		
	HR	95% CI	P Value	HR	95% CI	P Value
T stage	1.69	1.13–2.53	.01	2.69	0.99–7.34	.05
Abutting the optic disc	1.58	1.07–20.34	.02	1.61	1.06–2.44	.03
Initial metastasis	9.74	3.94–24.04	<.01	2.69	0.00–7.34	.05
Metastasis during follow-up	6.16	4.27–8.88	<.01	6.78	4.60–1.00	<.01
Older age	1.04	1.02–1.05	<.01	1.04	1.02–1.05	<.01

TABLE 7. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Factors Predictive for Metastasis

	Univariate Analysis			Multivariate Analysis		
	HR	95% CI	P Value	HR	95% CI	P Value
Abutting the optic disc	1.48	1.04–2.17	.048	1.86	1.25–2.78	<.01
Extrascleral extension	1.90	1.09–3.30	.02	1.82	1.03–3.72	.046
Ciliary body extension	1.89	1.34–2.65	<.01	1.90	1.31–2.75	<.01
Local recurrence	1.26	1.05–1.50	.01	1.20	0.99–1.43	.05

TABLE 8. Proton Beam Therapy for Large Choroidal Melanomas in 492 Consecutive Cases: Data on Large Choroidal Melanoma Studies Compared With Other Published Studies

	Number of patients	Mean tumor thickness mm (range)	Mean basal diameter mm (range)	Treatment type	NVG rate	Enucleation rate	Retinopathy and neuropathy rate	Tumor local control rate
Present study	492	8.8 (2–15)	14.9 (7–24.1)	PBRT	27%	19.5%	23.4%	94%
Semenova et al ¹⁵	47	8.6 (3.8–12.3)	15.8 (12.1–19.9)	¹⁰³ P	17%	11%	8.3%	91%
Shields et al ²³	354	9.0 (9.8–16.0)	14 (5.0–20.0)	¹²⁵ I	21%	24%	66%	91%
Fuss et al ²⁶	78	6.4	12.2	PBRT	17.9%	15.4%	51%	90.5%
Mosci et al ¹¹	132	9.8 (6.2–13.5)	15.2 (8.3–21.3)	PBRT	-	26%	43%	86%
Conway et al ²⁵	21	8.6	18.7	PBRT	38.1%	46%	24%	67%
Bechrakis et al ²⁴	152	9.0	14.6	¹²⁵ I	33%	6%	9.5%	88.8%

¹²⁵I = iodine 125; ¹⁰³P = palladium 103; PBRT = proton beam radiotherapy.

retinopathy of 9.5%, neuropathy of 9.5%, and cataract of 28.6%. Knowing that our patients tended to have tumor closer to the optic nerve than those in brachytherapy studies, such historical comparisons may only be hypothesis-generating in showing fewer posterior radiation complications with PBT than brachytherapy for large melanomas.

Another major complication of radiation treatments is NVG as NVG can compromise ocular preservation. Our crude rate of NVG was quite important, as it reached

27.0% with a mean follow-up of about 5 years. In comparison, Shields' study¹⁶ found a 21.0% 5-year rate of NVG and Semenova's¹⁵ 17.0%. Bechrakis et al²⁴ found 6.0% with plaque brachytherapy and transcleral resection. In Fossand associates' PBT study,³¹ 34% of the 127 large melanoma patients developed rubeosis. In Char and associates' study,³² 34.9% of the patients had NVG after helium therapy. Although the risk of NVG may appear slightly increased in patients treated with PBT, compared to plaque brachytherapy,²³ our rate (27.0%) was one of the lowest

following PBT. This might be due, at least partially, to the use of intravitreal VEGF inhibitors for the treatment of NVG after 2008. This was despite increasing rates of NVG after 2008, which probably reflects a trend toward irradiation of even larger tumors over time. However, it is interesting to note that enucleation rates decreased over the same period. This can be explained by a better management of complications.

Enucleation can be used either as uncontrolled NVG or as salvage treatment after tumor recurrence. In the COMS trial, 11% of the patients undergoing the brachytherapy arm ultimately underwent enucleation.⁵ Thus, an impressive proportion of patients could avoid up-front enucleation. In a recent study of small to large melanomas,³³ 19% of the patients underwent secondary enucleation after brachytherapy as a first-intent treatment in most patients. Secondary enucleation was performed for lack tumor response in 18%, tumor recurrence in 61%, and NVG in 21%. In Shields and associates' study of large choroidal melanoma, the 15-year rate of enucleation reached 41% after plaque radiotherapy,²³ and in Conway and associates' study²⁵ the rate was 46% after PBT for extra-large uveal melanomas. In our study, the enucleation rate was 19.5%, again suggesting a benefit of new surgical techniques such as endoresection and intravitreal therapeutics. While Egan and associates³⁴ identified involvement of the ciliary body as one of the risk factors for enucleation, several studies, like ours, have consistently identified radiation-induced retinopathy and post-PBRT retinal detachment as other major factors for enucleation. While NVG (30%) tends to be less frequent because of its better management, local recurrence (32%) might decrease in future years with optimized definition of the tumor borders, with more systematic use of ultrabiomicroscopy for ciliary body melanomas and wide-field retinographies for posterior pole tumors extending toward the equator.

Five-year local control rates were similar in other major studies and ours, with 91% in Shields and associates,²³ 91% in Fuss and associates,²⁶ 91% in Semanova and associates,¹⁵ 89% in Bechrakis and associates,²⁴ 86% in Mosci and associates,¹¹ and 94% in ours. In even larger tumors (mean diameter 18.7 mm), like in Conway's study,²⁵ the local control may drop to 67% at 2 years. Our mean tumor thickness and maximal tumor diameter were similar to the other large melanoma studies. Recurrence was primarily treated with enucleation (65.5%), but about one third of patients could still undergo PBT-based reirradiation. Tumor stage was associated with a significant risk

of enucleation, metastasis and poorer overall survival ($P = .05$). However, 5-year specific survival was still as high as 75%. This compares with 53% and 43% 5-year mortality rates in a meta-analysis³⁵ and the COMS⁸ after enucleation for large melanomas, respectively. Our strategy thus appears to yield favorable survival rates. Further, in consideration of the high mortality risk of large melanoma patients, it may be considered even more important to preserve the eye and its function.

Another conservative treatment approach, first described by Peyman and associates,^{36,37} is called endoresection, in which internal resection is performed via the pars plana using vitrectomy instruments. Endoresection is performed following PBRT of large choroidal melanoma to prevent the risk of radiation-induced retinal detachment and neovascular glaucoma, which represent the toxic tumor syndrome.³⁸ New therapies such as intravitreal antiangiogenic inhibitors will allow us to limit the toxic tumor syndrome in patients with large choroidal melanoma involving the ciliary body.³⁹ Exoresection is also used for patients who are unlikely to do well with the other principal forms of conservative treatment, but it requires a profound hypotensive anesthesia to reduce the risk of intraoperative hemorrhage.

Tumor size is one of the strongest risk factors for the loss of vision in irradiated eyes.^{40,41} In our series, BCVA at last visit was 20/200 or better in 20% of patients. In Shields and associates' plaque study about large choroidal melanoma,²³ 89% of their patients had $\leq 20/200$. Puusaari and associates²⁹ reported a 42% 1-year probability of having $\geq 20/200$ visual acuity after brachytherapy of large choroidal melanoma. Visual outcomes are heterogeneous among studies and correlate with tumor location and the distance to the macula and optic disc. Initial poor visual acuity is another important determinant of ultimate visual acuity that is itself related to tumor size and location,^{42,43} and in that sense our patients had unfavorable initial characteristics compared to other study patients.

Finally, proton beam therapy may be considered as a safe and efficient alternative to enucleation in patients with large choroidal melanomas (defined as AJCC stage T3 and T4) with respect to survival rates, local control, and visual acuity. With the development of such techniques as surgical endoresection and intravitreal injections or laser therapies, eyeball-threatening complications are increasingly manageable. PBT may thus increasingly be a treatment of choice in patients with large choroidal melanomas.

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